

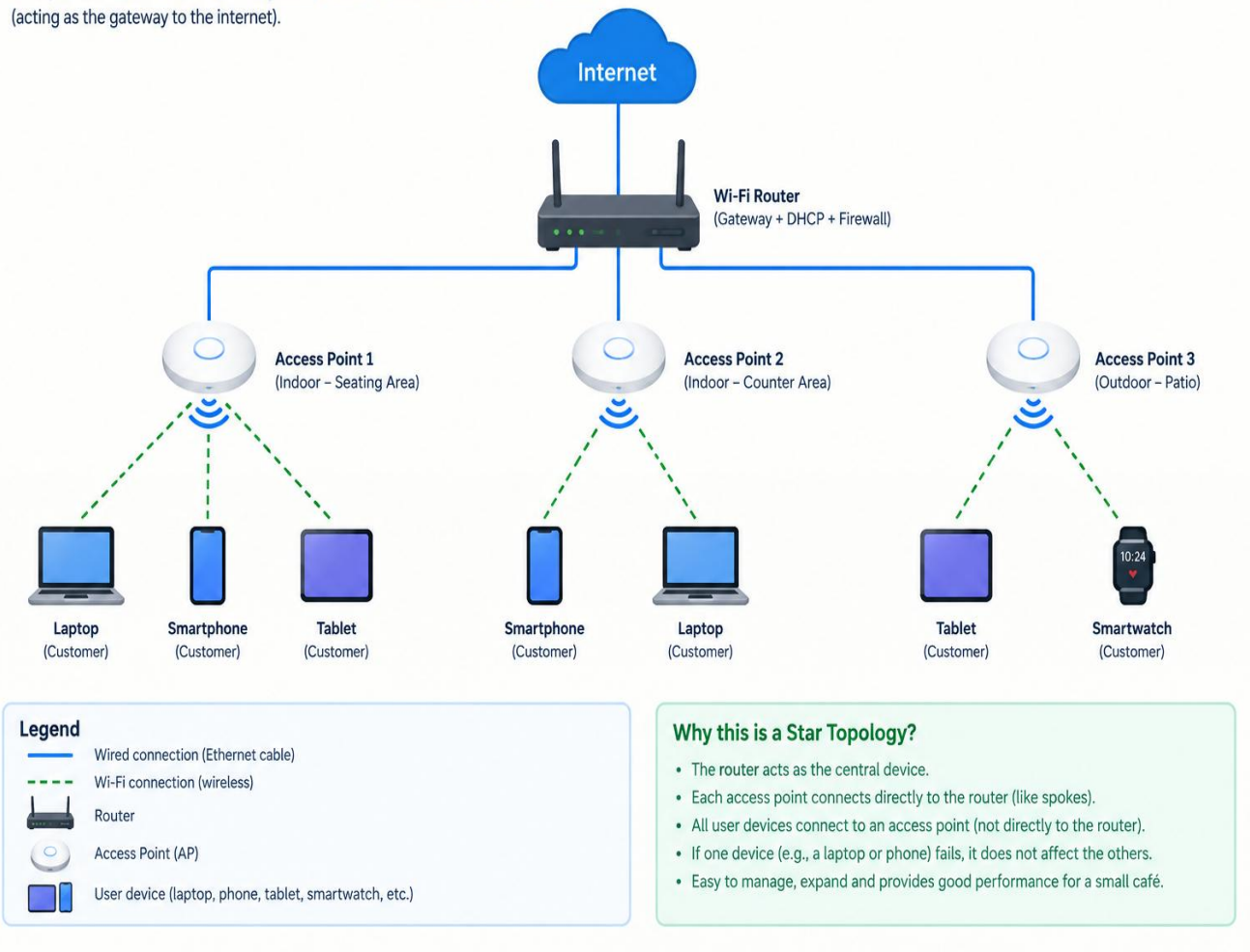
NETWORK TOPOLOGIES AND DESIGNS

1. Draw a network topology diagram for a small café Wi-Fi setup using the star topology, labeling all devices (router, access points, user devices) and connections clearl

->

Café Wi-Fi Network Topology (Star Topology)

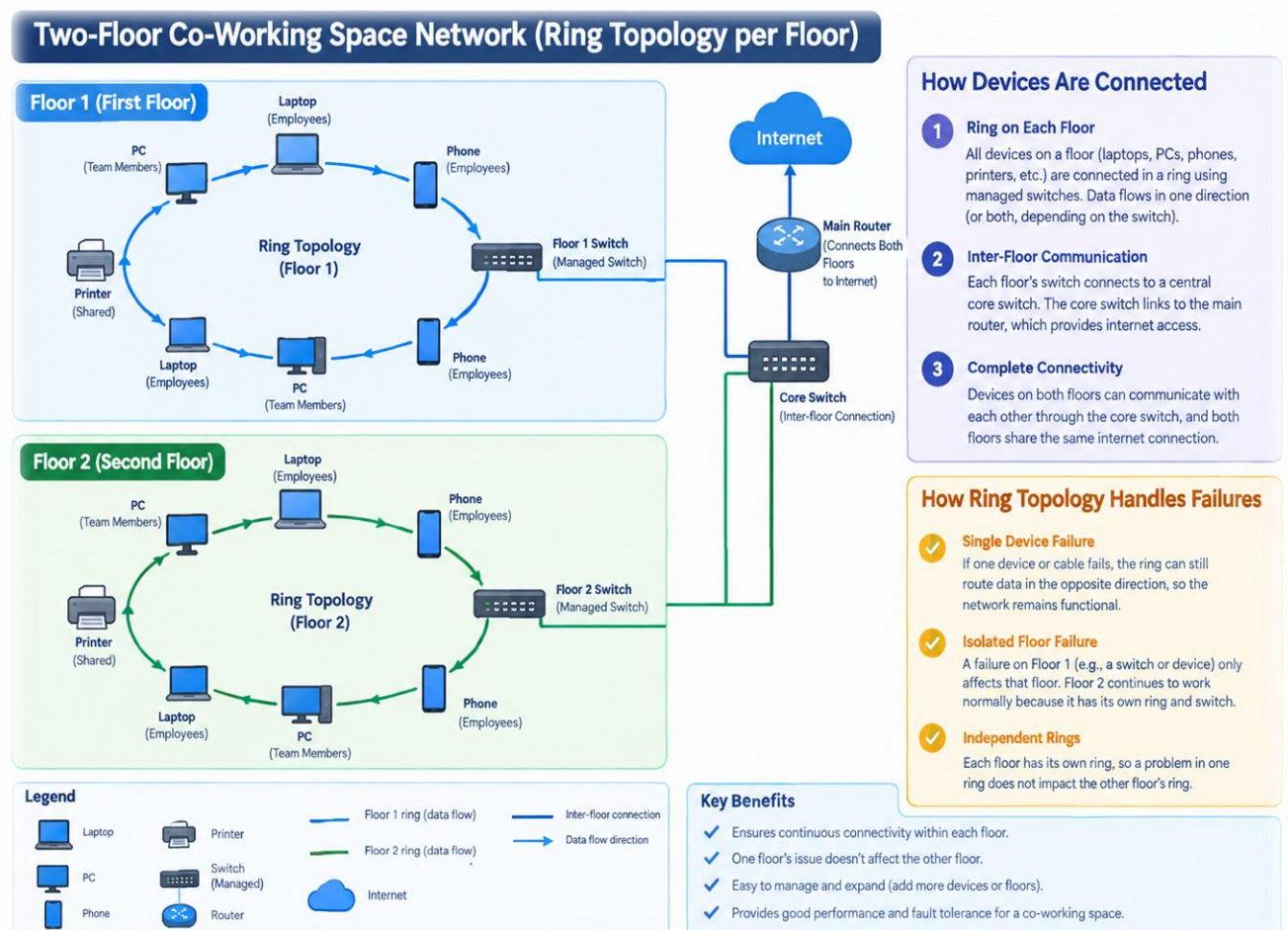
Multiple devices connect to access points, which are connected to a central router (acting as the gateway to the internet).



2. Design a network for a two-floor co-working space (like a mini WeWork) using a ring topology for each floor and explain how devices are connected to maintain communication between floors.

Hint: Show how the ring topology helps in handling device failures on one floor without affecting the other.

->



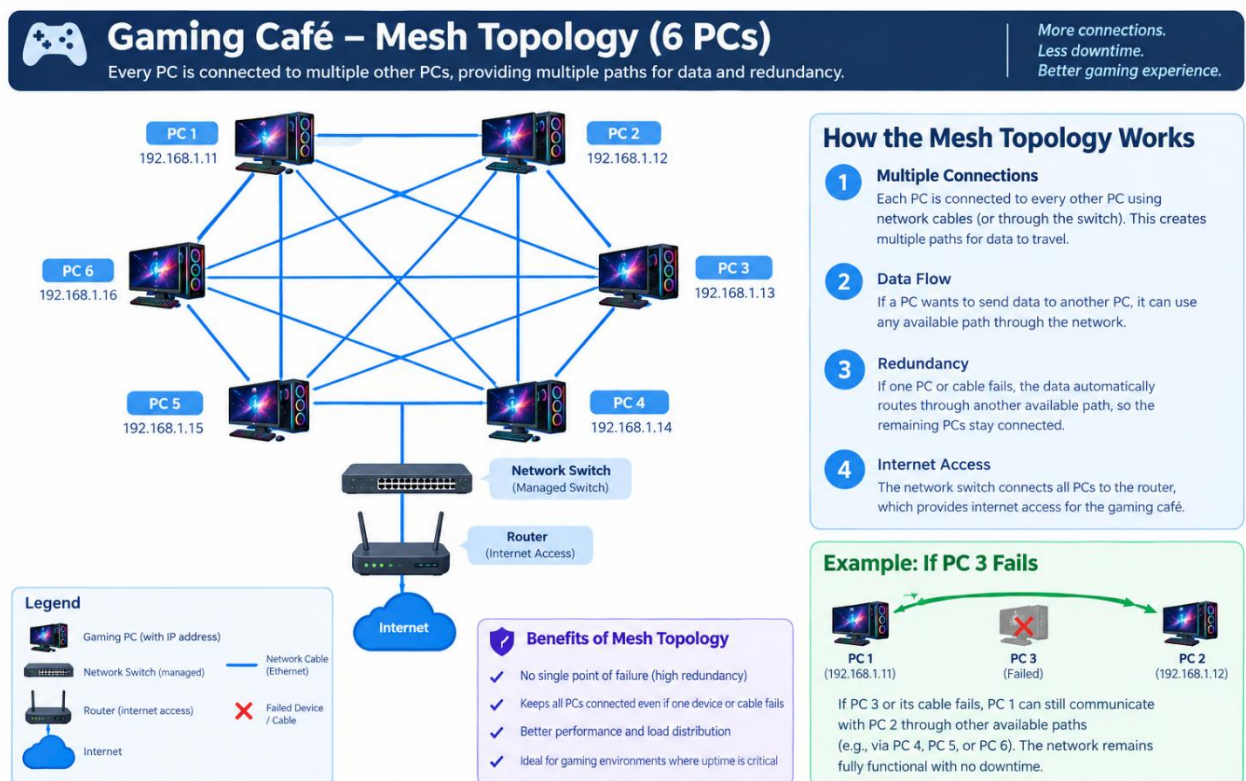
How communication works:

- Floor 1:** PCs, laptops, phones, and printers are connected in a ring.
- Floor 2:** A separate ring connects its PCs, laptops, phones, and printer.

3. **Between floors:** The Floor 1 and Floor 2 switches connect to the **core switch**, allowing devices on both floors to communicate.
4. **Internet access:** The core switch connects to the main router, which connects to the Internet.
5. **Failure handling:** If one device/link in a ring fails, data can be routed around the other direction of the ring. A problem isolated to Floor 1 does not directly disrupt the separate Floor 2 ring.

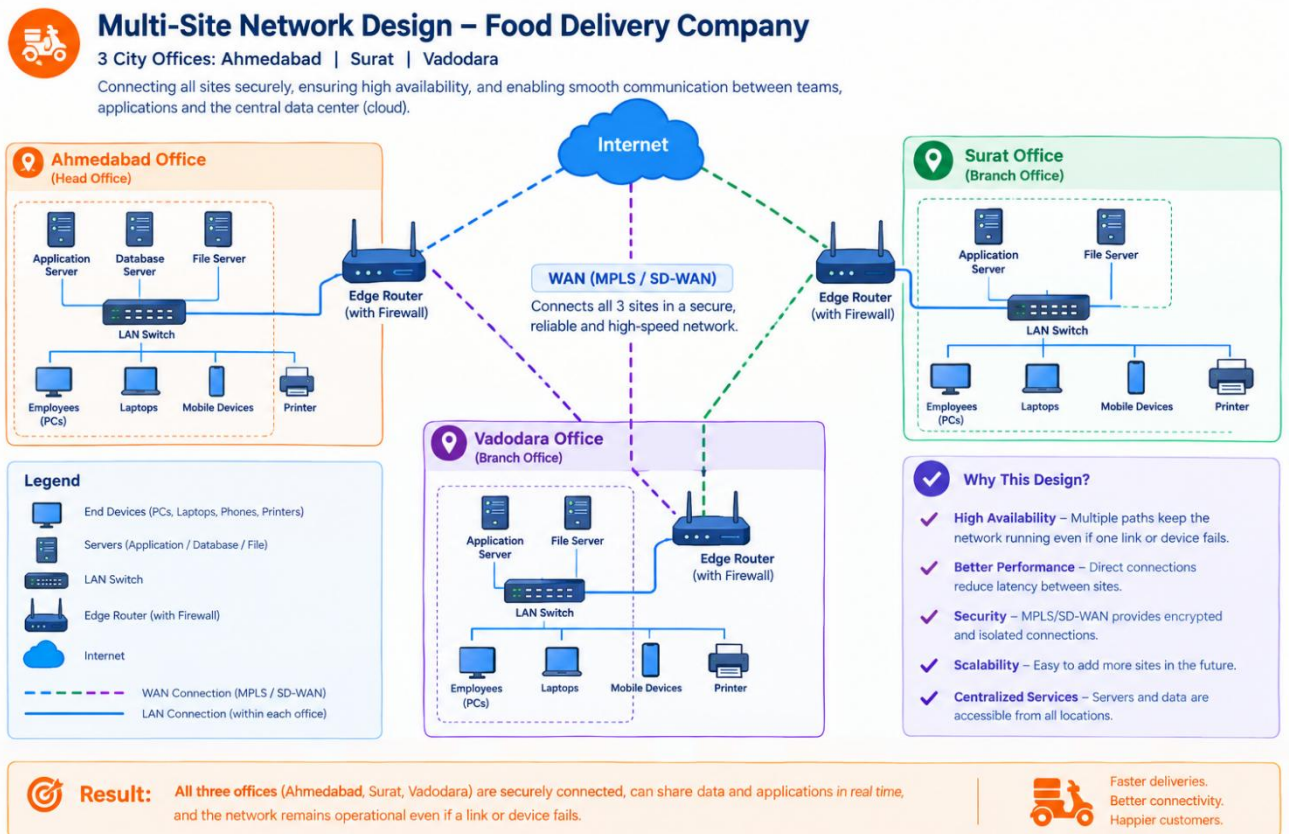
3. Create a mesh topology diagram for a gaming café with 6 PCs, showing how redundancy is achieved if any one PC or cable fails.

->



4. Plan a multi-site network for a food delivery company with 3 city offices (Ahmedabad, Surat, Vadodara) — show how you would connect the sites, what topology you would use between them, and briefly explain your choice.

->



Multi-Site Food Delivery Network

Recommended inter-site topology: Full Mesh WAN using MPLS/SD-WAN

- **Ahmedabad:** Head office with application, database, and file servers.
- **Surat:** Branch office with local application/file services.
- **Vadodara:** Branch office with local application/file services.
- Each office has an **edge router/firewall** connected to its local LAN switch.
- The three sites are connected through a **full-mesh WAN**, providing multiple communication paths.
- If one WAN link fails, traffic can use another available path, improving availability.
- **LAN topology:** Star topology within each office, which makes local management and expansion easier.
- **Why this design:** It supports secure inter-office communication, centralized applications/data, redundancy, and future expansion to additional cities.

5. Imagine your network needs to double in size next year (e.g., from 10 to 20 devices per site). Write a short expansion plan describing what changes you would make to your original topology to ensure scalability and minimal downtime.

Hint: Consider adding switches, routers, or changing topology types if needed.

-> **Network Expansion Plan: 10 to 20 Devices per Site**

If the food delivery company's network grows from **10 to 20 devices at each site**, I would expand the existing network without changing the overall WAN design.

Changes I would make

1. Add an additional managed switch

- Add a second 24-port managed switch at each office.
- Connect the new switch to the existing switch using an Ethernet uplink.
- This provides enough ports for new PCs, laptops, printers, phones, and other devices.

2. Keep the star topology inside each office

- Continue using a **star topology** because it is easy to manage and expand.
- New devices can simply be connected to the new switch rather than redesigning the whole network.

3. Upgrade routers if required

- Check whether the existing routers can handle the increased number of devices and traffic.
- If required, replace them with higher-capacity routers/firewalls supporting more simultaneous connections and faster WAN speeds.

4. Expand IP addressing

- Increase the DHCP address pool so all 20+ devices can receive IP addresses.
- Use VLANs such as **Staff, Guest Wi-Fi, and Servers** to organize and secure the network.

5. Maintain the inter-site WAN

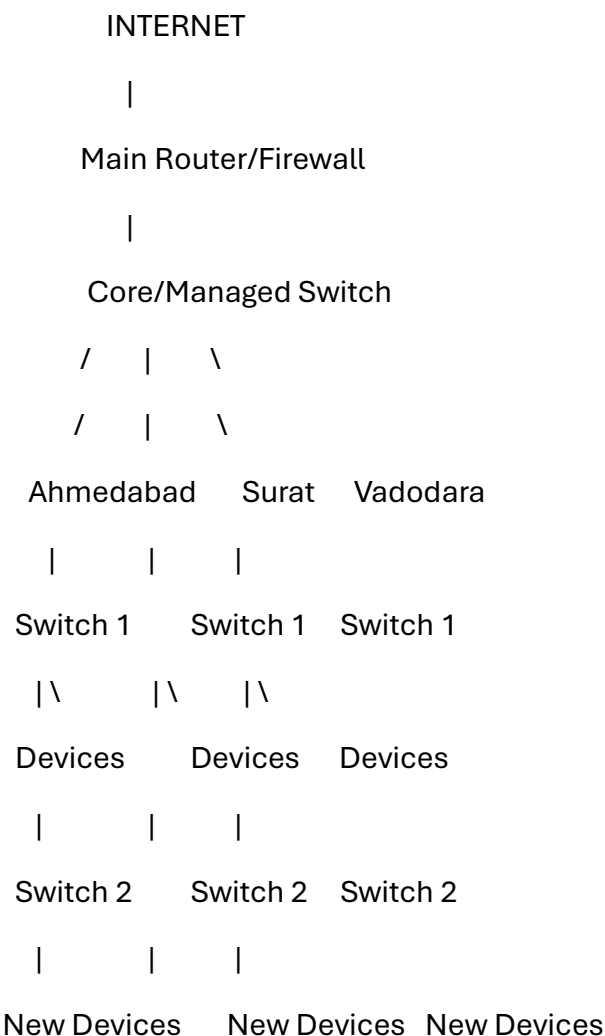
- Keep the **full-mesh MPLS/SD-WAN connection** between Ahmedabad, Surat, and Vadodara.

- If traffic increases significantly, upgrade the WAN bandwidth rather than replacing the topology.

6. Minimize downtime

- Install and configure the new switches before connecting them to the production network.
- Add devices and migrate connections during low-usage hours.
- Keep a backup configuration of routers and switches so the previous setup can be restored quickly.

Simple expanded topology



Result: Each site can grow from about **10 to 20+ devices** without replacing the entire network. The modular switch-based star topology makes expansion straightforward, while the existing inter-site WAN continues to provide communication between the three off